

# The AI-Enhanced Deliverable

**The interactive map is the AI enhancement of the Part 1 slide; I also used AI to build the site that delivers it**

## What this deliverable is

The exercise asks for AI-driven enhancements to the one-page slide from Part 1, in a format that can be embedded in a website. **The enhancement is the [interactive map](#)**: the slide’s static hero map, rebuilt with AI as an explorable Leaflet map — hover any hex for its plan-enabled units, toggle the 3,419 underlying opportunity parcels, turn the Mountain Line routes on and off, click any parcel for its details. It is a single self-contained HTML file, so it embeds in any web page — including a client-facing one — with one line of code.

AI was also used to build the delivery vehicle: this [Quarto](#) website, which packages the enhanced map together with the Part 1 slide ([download](#)), the full [data story](#), and the [methods audit trail](#) — so the deliverable arrives polished and is easy to transmit to clients for review. The site renders from a public repository, republishes automatically on every push, and every number regenerates from one command (`Rscript code/run_all.R`).

## How AI was used

The exercise asks for AI-driven enhancements to the Part 1 deliverable, and that is where I used it. I designed and built the analysis — the capacity framework, the opportunity screen and its exclusion rules, the hot-spot methodology, and the pipeline behind them. **Claude (Anthropic)** was then used to enhance and harden the deliverable:

- **Enhancing the slide’s hero map into the interactive map** — the Leaflet build, the hover and toggle behavior, and the self-contained packaging that makes it embeddable.
- **Building this website in Quarto** — the pages, navigation, theme, and publishing workflow that package all three parts into a polished deliverable a client can review from a single link.
- **Auditing the code** — reviewing the pipeline for correctness and reproducibility, and checking the cartography programmatically, including colorblind-safety validation (deuteranopia, protanopia, tritanopia) of the palette before it touched a map.

What stayed human: the choice of story (plan-enabled capacity, not another choropleth), the framing against the Growth Policy rather than zoning, the analysis and its assumptions, the exclusion rules and their ethics, the recommendations, and the final read on every number before it shipped.

## Why this transparency matters

A firm that publishes AI-polished work owes its clients clarity about where the analysis ends and the polish begins. Everything quantitative on this site traces to the reproducible pipeline in the [repository](#); the AI acceleration is documented here; and the assumptions that drive the headline numbers are inspectable in one file (`code/00_setup.R`). That is the standard I would bring to client work.